NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Electronics And Communication Engineering Department

# Digital Signal Processing Lab Equipment

| Equipment Overview | |
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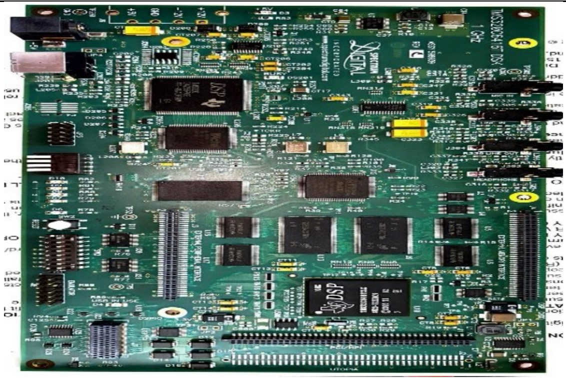
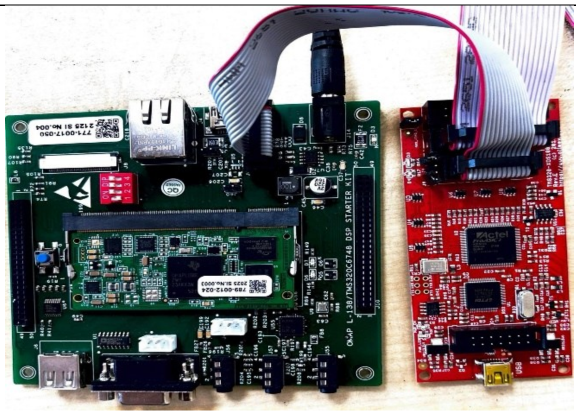
| **<u>DSP Starter KIT: TMS320C-6416T</u>**<br><br>The **TMS320C6416T DSP Starter Kit (DSK)** is a low-cost, all-in-one hardware and software development platform designed by Texas Instruments and Spectrum Digital. It allows engineers and researchers to prototype and evaluate high-performance digital signal processing applications based on the **TMS320C64x fixed-point DSP generation.** |  |
| **<u>DSP Starter KIT: TMS320C6748</u>**<br><br>The **TMS320C6748 DSP Development Kit**, commonly referred to as the **Low-Cost Development Kit (LCDK)**, is a high-performance evaluation platform designed by Texas Instruments. It serves as an accessible hardware and software sandbox for engineers, students, and researchers working on real-time signal processing, embedded analytics, biometrics, and high-fidelity audio engineering. |  |
| **<u>DSP Processor Trainer Kit: TM5320C6713</u>**<br>The **TMS320C6713 DSP Starter Kit (DSK)**—often referred to as the **TMS320C6713 Development Kit**—is a highly popular, low-cost evaluation platform designed by Texas Instruments and Spectrum Digital. It is built around a high-performance **floating-point digital signal processor**, making it an industry and academic standard for teaching real-time audio processing, filter design, and control systems |  |

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<td><u>Digital Signal Oscilloscope:</u><br>A Digital Storage Oscilloscope (DSO) is an essential electronic test instrument used to capture, visualize, analyze, and store electrical signals over time. Unlike older analog oscilloscopes that use a cathode-ray tube to display waveforms directly, a DSO converts electrical voltages into digital data using an Analog-to-Digital Converter (ADC) and displays them on a digital screen.</td>
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<td><u>Mixed Signal Oscilloscope:</u><br>A Mixed Signal Oscilloscope (MSO) is a hybrid electronic test instrument that combines the detailed analog visualization of a Mixed Domain or Digital Storage Oscilloscope (DSO) with the multi-channel digital timing capabilities of a logic analyzer. It allows engineers to view and analyze both analog waveforms and digital logic signals simultaneously on a single screen, perfectly synchronized to a single timebase.</td>
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<td><u>Waveform/Signal Generator:</u><br>A Waveform Generator (frequently called a Signal Generator or Function Generator) is an essential electronic test instrument used to inject precise, predictable electrical voltages into a circuit. While an oscilloscope listens to a circuit to display signals, a waveform generator talks to a circuit by sourcing controlled electrical stimuli to test its behavior, verify its design, or discover faults.</td>
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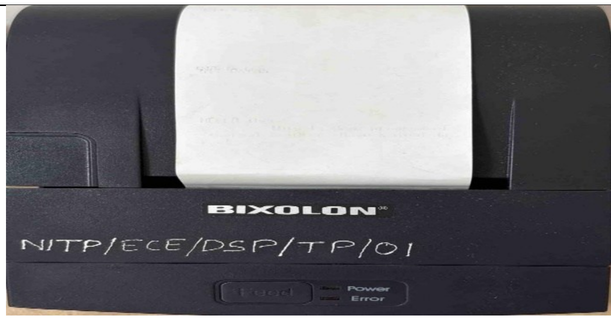
<td><u>Thermal Printer:</u><br>A Thermal Printer is a digital printing instrument that uses heat instead of liquid ink or toner ribbon to produce text and images on paper. Because they contain fewer moving parts and do not require ink cartridge refills, thermal printers are highly reliable, low-maintenance, and widely used in commercial, logistics, and retail spaces</td>
<td>
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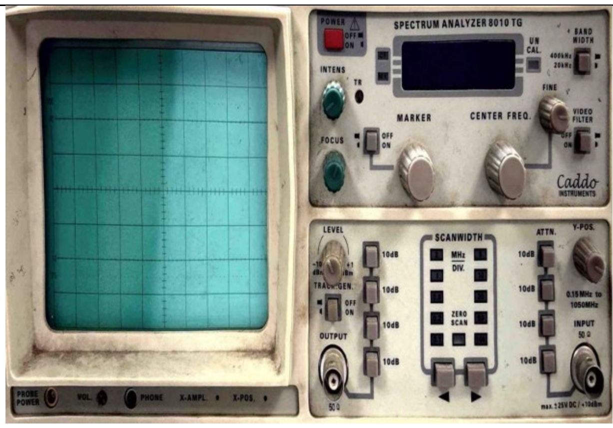
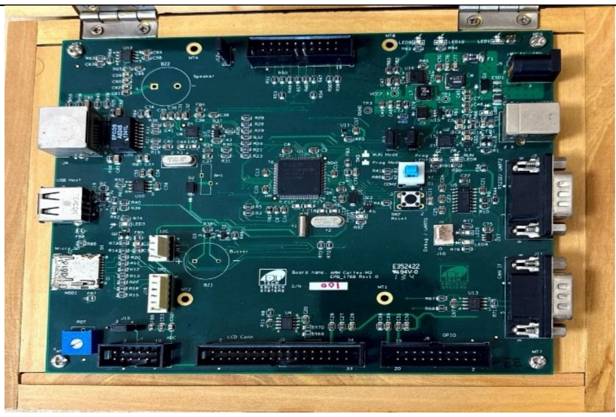
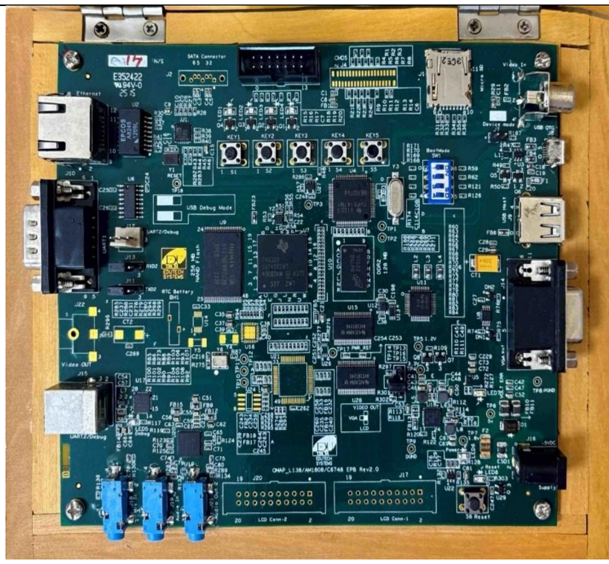
| **<u>Spectrum Analyzer:</u>**<br>A **Spectrum Analyzer** is a sophisticated electronic test instrument used to visualize and analyze the strength of electrical signals across a range of frequencies. While an oscilloscope displays signals in the **time domain** (Voltage vs. Time), a spectrum analyzer translates signals into the **frequency domain** (Amplitude vs. Frequency), allowing engineers to look at individual frequency components that make up a complex signal. |  |
| **<u>ARM Cortex M3 kit: LPC1768</u>**<br>The **LPC1768 evaluation and evaluation kit** is a popular hardware development platform built around an **ARM Cortex-M3 microcontroller** manufactured by NXP Semiconductors. It serves as a robust sandbox for engineers, embedded students, and firmware developers to prototype and evaluate real-time applications such as industrial automation, automotive electronics, and connected IoT systems. |  |
| **<u>Educational Practice board for TMS320C6748:</u>**<br>The **Educational Practice Board (EPB) for the TMS320C6748** is a specialized, all-in-one laboratory trainer kit designed for universities, technical institutions, and engineering labs. Built as an educational alternative to standard industrial hardware, this platform expands on the native <u>Texas Instruments TMS320C6748 floating-point DSP architecture</u> by populating the board with heavy-duty user interfaces, multimedia connectors, and diagnostic tools to help students master real-time signal, audio, and image processing. |  |

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**Finger Print Sensor interfacing kit:**

A **Fingerprint Sensor Interfacing Kit** is a hardware development bundle designed to teach developers, students, and engineers how to integrate biometric authentication into embedded applications. The kit centers around a self-contained biometric module (such as the widely used R307 Optical Sensor or AS608 Module) paired with a host microcontroller development board.

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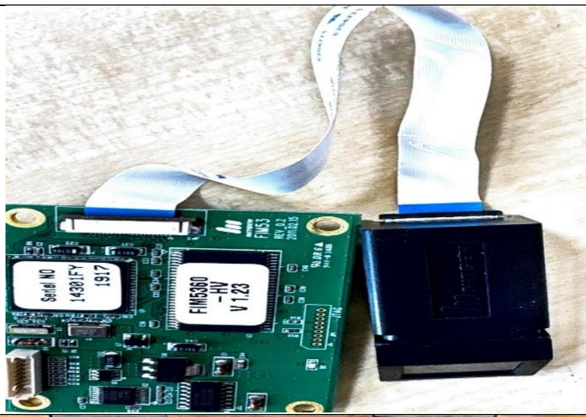


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**General Purpose IO Board:**

A **General Purpose Input/Output (GPIO) Board** (frequently called a **GPIO Expansion Board** or **Breakout Board**) is a hardware accessory designed to extend, protect, and distribute the digital control pins of a primary computing host like a microcontroller, Single Board Computer (SBC), or automated industrial controller [sparkfun.com]. It bridges sensitive processing cores with real-world physical electronics like switches, sensors, relays, and indicators.

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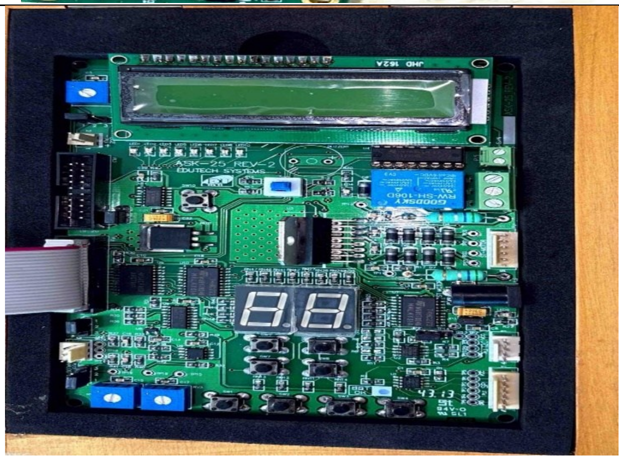


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**RFID Interfacing Kit:**

An **RFID (Radio Frequency Identification) Interfacing Kit** is a hardware development bundle designed to teach engineers, students, and makers how to implement wireless tracking, access control, and asset identification systems. The kit typically pairs a popular RFID reader module—such as the low-cost **RC522 (13.56 MHz HF)** or **EM-18 (125 kHz LF)**—with a host microcontroller and a variety of transponder tags.

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## WIFI Module: RN174

The **RN174 WiFly GSX Super Module** (marketed under Microchip Technology following their acquisition of Rving Networks) is a field-ready, Wi-Fi Alliance-certified **802.11 b/g wireless prototyping board**. Rather than being just a bare IC, the RN174 serves as a comprehensive **development evaluation platform built around the core RN-171 Wi-Fi module**, designed specifically to add networking capabilities to low-power embedded designs without forcing engineers to rewrite extensive hardware


## Zigbee interfacing kit:

A **Zigbee Interfacing Kit** is a hardware development bundle designed to teach developers, engineers, and students how to implement low-power, short-range **wireless mesh networking**. The kit typically pairs popular Zigbee modules—such as the industry-standard **Digi XBee (Series 2 or 3)** or **TI CC2530**—with interfacing breakout boards, transceivers, and host microcontrollers.